problem1

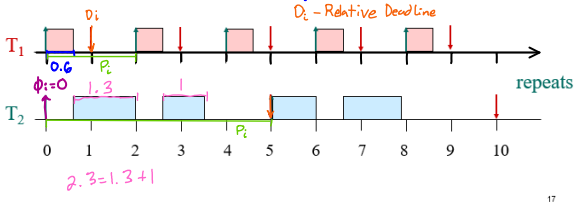
Sunday, February 1, 2026 8:45 PM

Proof of Non-tightness

To see that $\Delta > 1$ doesn't imply non-schedulability, consider the following example.

Example: We have two tasks $T_1 = (2, 0.6, 1)$ and $T_2 = (5, 2.3)$. $\Delta = 0.6/1 + 2.3/5 = 1.06$. Nonetheless, we can schedule this task set under EDF:

P_i – Period
e_i – Execution Cost
D_i – Relative Deadline



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Notation of Real-Time Tasks

- Model assumed in this chapter:
  - n periodic tasks $T_1,\ldots,T_n$.
  - The "rest of the world" periodic model is assumed.
  - T_i is specified by (ϕ_i, p_i, e_i, D_i), where
    - ϕ_i is its phase (0 unless indicated)
    - p_i is its period (an integer or, at least, rational)
    - e_i is its execution cost per job, and
    - D_i is its relative deadline.
    - Will abbreviate as (p_i, e_i, D_i) if $\phi_i=0$, and (p_i, e_i) if $\phi_i=0$ & $p_i=D_i$.

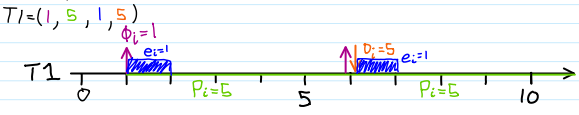
And
Not XOR

1. (10 points) Determine if the task set is schedulable according to (a) deadline monotone scheduling under the system utilization test and then (b) GTDA (Generalized Timing Demand Analysis), but this time with the priority assignment in the listed order (T1,T3,T2). Also calculate the in-phase busy period. T1 = (7, 1), T2 = (12, 5), T3 = (9, 3, 11)

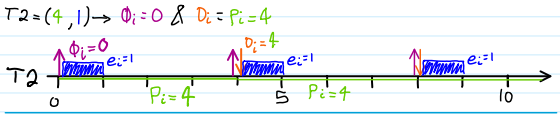
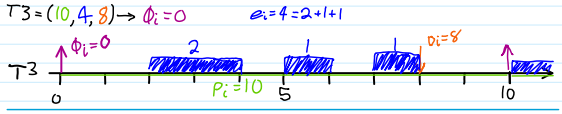
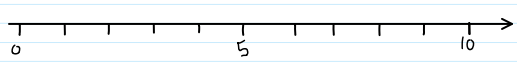
Turn in file problem1.txt.

ϕ_i – Phase – Time after 0 Arrow before task starts to execution (Orange Arrow)
P_i – Period – Duration between two Purple Arrows
e_i – Execution Cost – Task Duration
D_i – Relative Deadline – Time after Purple Arrow where Orange Arrow should be Placed

$T=(\phi_i, P_i, e_i, D_i)$
$T=(P_i, e_i, D_i) \rightarrow \phi_i=0$
$T=(P_i, e_i) \rightarrow \phi_i=0$ & $D_i=P_i$
$T1=(1, 5, 1, 5)$



$T2=(4, 1) \rightarrow \phi_i=0$ & $D_i=P_i=4$



$T3=(10, 4, 8) \rightarrow \phi_i=0$ $e_i=4=2+1+1$